



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.6C

Evaluate Expressions

Lesson 6-2 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can evaluate algebraic expressions by substituting values for the variables.

Language Objective: I can explain my steps using the words variable, expression, evaluate, and substitute.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Variable

Expression

Evaluate

Substitute

Coefficient

Like terms



Tap any word to see what it means and a picture.

1

A letter that stands for an unknown number, like x , is a .

2

A group of numbers, variables, and operations with no equal sign is an

.

3

To find the value of an expression for given values is to it.

4

To replace a variable with a number is to .

5

A number multiplied by a variable, like the 4 in $4y$, is a .

6

Terms with the same variable part, like $2x$ and $6x$, are .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

When you evaluate an expression like $4x^2 + 3$, what order do you follow, and why must you square before you multiply?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐

Variable — A letter that stands for an unknown number.

Variable — Una letra que representa un número desconocido.

☐

Expression — A math phrase with numbers and letters, but no equal sign.

Expresión — Una frase matemática con números y letras, pero sin signo igual.

☐

Evaluate — To find the value by putting numbers in for the letters.

Evaluar — Encontrar el valor poniendo números en lugar de las letras.

☐

Substitute — To put a number in place of a letter.

Sustituir — Poner un número en lugar de una letra.

☐

Coefficient — The number in front of a letter, like the 3 in $3x$.

Coeficiente — El número frente a una letra, como el 3 en $3x$.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

First I substitute, then I do ____ because of the order of operations.

Primero sustituyo, luego hago ____ por el orden de las operaciones.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Substitute the number of speakers for s , then evaluate.

Evidence: Students substitute $s = 5$ ($150 + 12 \times 5 = 210$), explain 150 is the fixed venue fee and 12 is the cost per speaker, and note s can be any number.

Reasoning: This evidence connects the definition of expression to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **First I substitute, then I do ____ because of the order of operations.**

Primero sustituyo, luego hago ____ por el orden de las operaciones.

- **I must square before multiplying because ____.**

Debo elevar al cuadrado antes de multiplicar porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Variable
2. **Notes 2:** Expression
3. **Notes 3:** Evaluate
4. **Notes 4:** Substitute
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **Watch for this mistake:** *A common mistake in Evaluate Expressions is treating a coefficient like an addend instead of a factor — for example, evaluating $3x + 7$ when $x = 4$ as $3 + 4 + 7 = 14$ instead of the correct $3(4) + 7 = 12 + 7 = 19$. A number written right next to a variable always means multiply, never add. Before you submit, ask: "Did I multiply the coefficient by the value I substituted, or did I accidentally add it?"*
8. **Try It 1:** A. 16 — Substitute $n = 5$: $2(5 + 3)$. Do the parentheses first: $5 + 3 = 8$. Then multiply: $2 \times 8 = 16$.
9. **Try It 2:** A. 36 — x^2 means x times itself. Substitute $x = 6$: $6^2 = 6 \times 6 = 36$.